

An exact fixed-chaos slice of the L^2 Haar-multishift commutant

Candidate partial result for arXiv:2005.04648

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Status. Candidate partial result, likely valid. We obtain an exact norm and boundedness criterion for a generator in any one fixed Haar-chaos order. The unrestricted commutant problem is *not* claimed.

1. Source question and result

Let h be the first Haar function and V_0, V_1 the two dyadic dilation operators. For a mean-zero function f , Astashkin and Terekhin define

$$T_f V^\alpha h = V^\alpha f.$$

The source notes that T_f commutes with the Haar multishift and states that a comprehensive description of its commutant is open even in L^2 [1]. It completely treats first-order Haar chaoses. The excerpt below records the open problem.

We solve the boundedness problem when f belongs to one prescribed Haar chaos. Use the source parametrization

$$f = \sum_{k_1, \dots, k_d \geq 0} \xi_{k_1, \dots, k_d} V_0^{k_1} V_1^{k_2} \cdots V_1 V_0^{k_d} h \quad (1)$$

and its symbol

$$\widehat{f}(z_1, \dots, z_d) = \sum_{k_1, \dots, k_d \geq 0} \xi_{k_1, \dots, k_d} z_1^{k_1} \cdots z_d^{k_d}.$$

Put $R = 2^{-1/2}$, and use the convention $H^2(\mathbb{D}_R^0) = \mathbb{C}$.

Theorem 1 (fixed-chaos commutant formula). *Let $d \geq 1$ and let $f \in \mathcal{H}_{ch}^d \cap L^2$ have the expansion (1). The densely defined operator T_f extends boundedly to the mean-zero subspace $L_0^2[0, 1]$ if and only if*

$$\sup_{|w| < R} \|\widehat{f}(w, \cdot)\|_{H^2(\mathbb{D}_R^{d-1})} < \infty. \quad (2)$$

In that case its exact norm is

$$\|T_f\|_{L_0^2 \rightarrow L_0^2} = 2^{-(d-1)/2} \sup_{|w| < R} \|\widehat{f}(w, \cdot)\|_{H^2(\mathbb{D}_R^{d-1})}.$$

(3)

For $d = 1$, this is exactly the scalar $H^\infty(\mathbb{D}_R)$ formula proved in the source.

$\{2^{|\alpha|/2}V^\alpha h\}_{\alpha \in \mathbb{A}}$ is the corresponding orthonormal basis. By this reason, the pair of operators V_0 and V_1 will be called the *Haar multishift*.

Given a function f , we define the linear operator T_f by

$$T_f V^\alpha h = V^\alpha f, \quad \alpha \in \mathbb{A}.$$

In what follows, we repeatedly use the fact that the operator T_f commutes with the operators V_0 and V_1 , that is, for each mean zero Haar polynomial x we have

$$TV_0 x = V_0 T x \quad \text{and} \quad TV_1 x = V_1 T x.$$

Conversely, every operator T , defined on the linear manifold of all mean zero Haar polynomials and commuting with V_0 and V_1 , can be represented as T_f , where $f := Th$ (indeed, $Th_\alpha = TV^\alpha h = V^\alpha Th = f_\alpha$, $\alpha \in \mathbb{A}$).

Of fundamental importance is the problem of a description of the commutant $c(V_0, V_1)_{L^p}$ of the Haar multishift, which is defined as the set of mean zero functions $f \in L^p$ such that the operator T_f is bounded on L^p . More precisely, we say that the operator T_f is bounded on the space L^p if it is bounded on the subspace of L^p of all mean zero functions; by $\|T_f\|_{L^p}$ we will denote its usual operator norm on this subspace.

Recall that every operator T commuting with the shift operator S in the Hardy space $H^2(\mathbb{D})$ has the form T_u , where $T_u v(z) = u(z)v(z)$, and it is bounded in $H^2(\mathbb{D})$ if and only if $u \in H^\infty(\mathbb{D})$ (see e.g. [10, Chapter 17, Problem 147]). In contrast to this classical situation, a comprehensive description of the commutant of the Haar multishift is still an open problem even in the L^2 -case. On the one hand, the boundedness and even the continuity of a function f , in general, does not guarantee the boundedness of the operator T_f in L^2 [20]. On the other hand, T_f is bounded in the latter space under a certain minimal smoothness condition [27]. Concerning the case of a function $f \in \mathcal{H}_{ch}^1$, the condition $f \in L^\infty$ is sufficient but not necessary for the operator T_f to be bounded on L^p for each $1 < p < \infty$ [2, Theorem 7].

It is clear that the system of dilations and translations of a function f forms a basis in the space L^p equivalent to the Haar system if and only if the operator T_f is an isomorphism of L^p . Moreover, as is easy to see, the inverse operator T_f^{-1} commutes with the Haar multishift as well and therefore $T_f^{-1} = T_g$, where $g = T_f^{-1}h$.

Figure 1: Definition of T_f and the open L^2 commutant problem on page 7 of arXiv:2005.04648.

2. Proof intuition

In the normalized Haar basis, the two multishift generators are the left creation operators on binary words, whereas every commuting T_f appends the coefficient word on the right. A word in one fixed chaos order has a distinguished initial run of zeros. Removing that run leaves a terminal word from a suffix-free family, so the corresponding right-creation ranges are orthogonal.

The only remaining interaction is convolution along the initial zero run. That interaction is an ordinary unilateral shift. Its Wold model converts T_f into multiplication by an ℓ^2 -valued analytic function: the ℓ^2 coordinates are precisely the remaining $d - 1$ chaos variables. The multiplier norm is therefore an H^∞ norm in the first variable and an H^2 norm in all the others. This orthogonality is lost as soon as different chaos orders are mixed.

3. Free-word reduction

Let $\mathbb{F}_2^+$ be the set of finite binary words, including the empty word, with concatenation denoted by juxtaposition. The normalized functions

$$e_\alpha = 2^{|\alpha|/2} V^\alpha h, \quad \alpha \in \mathbb{F}_2^+,$$

form an orthonormal basis of L_0^2 . If $f = \sum_{\beta \in \mathbb{F}_2^+} a_\beta e_\beta$, then a direct calculation gives

$$T_f e_\alpha = \sum_{\beta \in \mathbb{F}_2^+} a_\beta e_{\alpha\beta}. \quad (4)$$

Thus the coefficient word is appended on the right. Define the right creation isometry $R_\beta e_\alpha = e_{\alpha\beta}$.

For $d > 1$, every word occurring in (1) has a unique factorization

$$\beta = 0^{k_1} \gamma, \quad \gamma = 1, 0^{k_2} \dots 1, 0^{k_d}.$$

Let Γ_d be the family of these γ ; put $\Gamma_1 = \{\emptyset\}$. The family Γ_d is suffix-free. Indeed, if γ were a proper suffix of δ , the deleted nonempty prefix of δ would begin with 1. It would contribute an additional one, although both words have exactly $d - 1$ ones. This is impossible.

Consequently the ranges of R_γ , $\gamma \in \Gamma_d$, are pairwise orthogonal: a word ending in two distinct terminal words would force one of them to be a suffix of the other. Write

$$b_{k,\gamma} = a_{0^k \gamma}, \quad \phi_\gamma(z) = \sum_{k \geq 0} b_{k,\gamma} z^k.$$

Equation (4) now becomes

$$T_f = \sum_{\gamma \in \Gamma_d} R_\gamma \phi_\gamma(R_0), \quad \|T_f x\|_2^2 = \sum_{\gamma \in \Gamma_d} \|\phi_\gamma(R_0)x\|_2^2. \quad (5)$$

4. Vector-valued Hardy multiplier

We record the functional-analytic step, including the infinite sum.

Lemma 1. *Let S be a pure isometry on a Hilbert space, and let $\Phi = (\phi_j)_{j \in J}$ be an $\ell^2(J)$ -valued analytic function whose Taylor coefficients are square summable. On the algebraic Wold span set*

$$Cx = (\phi_j(S)x)_{j \in J}.$$

Then C is bounded if and only if $\Phi \in H^\infty(\mathbb{D}; \ell^2(J))$, and

$$\|C\| = \sup_{|z| < 1} \left(\sum_{j \in J} |\phi_j(z)|^2 \right)^{1/2}.$$

Proof. The Wold unitary identifies S with multiplication by z on $H^2(\mathbb{D}; \mathcal{W})$, where $\mathcal{W} = \ker S^*$. Under this unitary, C is pointwise multiplication

$$g(z) \mapsto \Phi(z) \otimes g(z)$$

from $H^2(\mathbb{D}; \mathcal{W})$ to $H^2(\mathbb{D}; \ell^2(J) \otimes \mathcal{W})$. Boundary integration gives the upper estimate by $\|\Phi\|_{H^\infty}$, and normalized reproducing kernels give the reverse estimate at every point of the disk. For infinite J , apply this argument first to finite coordinate projections. Their squared pointwise norms increase to $\|\Phi(z)\|_{\ell^2}^2$, which proves both necessity and the asserted norm identity. $\square$

The isometry R_0 in (5) is pure: its wandering space is spanned by the vacuum and by words ending in 1. Orthogonality of the R_γ makes T_f isometrically equivalent to the column in Lemma 1. Hence

$$\|T_f\| = \sup_{|z|<1} \left(\sum_{\gamma \in \Gamma_d} |\phi_\gamma(z)|^2 \right)^{1/2}. \quad (6)$$

It remains only to translate the normalization. For $\gamma = 1, 0^{k_2} \dots 1, 0^{k_d}$, the word in (1) has length $k_1 + \dots + k_d + d - 1$, so

$$\phi_\gamma(z) = 2^{-(d-1+k_2+\dots+k_d)/2} \sum_{k_1 \geq 0} \xi_{k_1, \dots, k_d}(z/\sqrt{2})^{k_1}.$$

Putting $w = z/\sqrt{2}$ in (6), and using the coefficient norm of $H^2(\mathbb{D}_R^{d-1})$, gives

$$\sum_{\gamma \in \Gamma_d} |\phi_\gamma(z)|^2 = 2^{-(d-1)} \|\hat{f}(w, \cdot)\|_{H^2(\mathbb{D}_R^{d-1})}^2.$$

This proves Theorem 1.

5. Strictness, computation, and limitation

Condition (2) is stronger than $f \in L^2$. For example, fix all tail indices $k_2 = \dots = k_d = 0$ and choose the first-variable symbol

$$g(w) = (1 - \sqrt{2}w)^{-1/4}.$$

Then $g \in H^2(\mathbb{D}_R)$ but $g \notin H^\infty(\mathbb{D}_R)$. The corresponding order- d chaos belongs to L^2 , while Theorem 1 says that T_f is unbounded. At the opposite normalization extreme, $f = V_1 h$ gives $\|T_f\| = 2^{-1/2}$, as it must.

The accompanying finite-word audit uses an order-three polynomial with four terminal suffixes. The normalized vector norm and independently scaled mixed norm agree to twelve decimal places, while exact compression norms through depth eight increase toward the predicted value.

For $d > 1$, every output word in (4) has at least $d - 1$ more ones than the input. In particular, T_f is not onto and cannot produce a Haar-equivalent affine basis. More importantly, terminal families from different chaos orders are nested rather than suffix-free. Their ranges need not be orthogonal, and the Pythagorean reduction (5) breaks down. This is why the theorem is a proper slice of, not a full solution to, the source's unrestricted commutant problem.

6. Novelty and review status

The earlier paper [2] treats Haar chaoses of arbitrary fixed order and characterizes simultaneous boundedness in every L^p , while its exact continuous-invertibility result is for order one. The present source proves the single- L^2 scalar formula for order one. A bounded search using the exact titles together with fixed-chaos, commutant, and mixed-Hardy terms found no explicit formula (3). Since the abstract free-semigroup commutant model is standard, novelty is claimed only for the explicit fixed-Haar-chaos reduction and exact mixed norm, pending specialist review.

Human-review focus. Check the orientation in (4), the suffix-free argument, the infinite-coordinate passage in Lemma 1, and the factor $2^{-(d-1)/2}$ in (3).

References

- [1] S. V. Astashkin and P. A. Terekhin, *Sequences of dilations and translations equivalent to the Haar system in L^p -spaces*, arXiv:2005.04648.
- [2] S. V. Astashkin and P. A. Terekhin, *Sequences of dilations and translations in function spaces*, J. Math. Anal. Appl. **457** (2018), 645–671.